

Introduction into Computational Linguistics: Final Project Report

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Project Description

In my final project, I am going to compare the performance of two language models on machine translation downstream tasks with the Belarusian (BE) and the English (EN) languages.

The downstream tasks themselves are presented in three different sizes: The translation in both directions (BE to EN, EN to BE) of one sentence, of a three sentence set and the validation set, printing out only 15 segments in total.

My comparison will consist of:

1. Evaluating Model 1 against Model 2 without fine-tuning on the dataset.
2. Fine-tuning Model 2 with dataset and evaluating it.
3. Comparison of its performance with both the first model and its initial, non-fine-tuned state.

The project has followed the further workflow:

1. Loading Models
2. Loading Metrics (Google BLEU, SacreBLEU)
3. Loading Dataset
4. Evaluating the Two Models' Performance in Zero-Shot Manner
5. Fine-tuning Model 2
6. Evaluating Model 2 Performance in Zero-Shot Manner
7. Quantitative and Qualitative Result Analysis
8. Visualization of Hidden States of Model 2 in TensorFlow

Models' Description

The first model is **WelfCrozzo/T5-L128-belarusian**, a language model specifically trained on translation tasks including this language pair. Therefore, it can be used in zero-shot manner without fine-tuning. According to the model description, the translation downstream task of is preformed using specific tokens, such as <extra_id_6> for translation EN to BE and <extra_id_2> for translation BE to EN.

The second model is **google/mt5-small**, a pretrained language model that supports 101 languages but not trained on translation downstream tasks specifically. Thanks to the help of my professor, it has been proved that google mt5 cannot be used in the zero-shot manner as the previous model and, therefore, needs a proper fine-tuning.

I have not found any specifications about the specific tokens to use in Model 2 for selecting the language pair and the translation direction, so I am using a prompt describing the task before the text to translate.

Dataset Description

The dataset is an English-centric multilingual corpus covering 100 languages, meaning that all training pairs include English on either the source or target side. For my specific language pair ("be-en"), the dataset includes 71.3k rows of translations in the form of a dictionary including both the source ("be") and the target segments ("en"). It does not have any additional labels and also includes some untranslated segments, i.e. both source and target segments were in English, or unparallelled segments like {"be": "@ title: menu", "en": "Set Read-only"}.

In the real translation scenario some of the untranslated segments could be untranslated either, e.g. names of organizations like here:

Nevertheless, I would have wished that they were not present in the dataset, since evaluating them with metrics I have chosen could lead to misleading scores.

In my project I have split the dataset into three parts in the following proportions: Training set included 2000 segments, while both validation and test sets - 200 segments each. For each of the models, I used different preprocessing methods of the dataset, preserving the number of segments.

For Model 1, I used this code to preprocess the sets, as I wanted to use them only in inference stage for evaluating its output and not for fine-tuning:

```
# Truncating dataset
def truncate(example):
    return {
        "source": example['translation']['be'],
        "translation": example['translation']['en']
    }

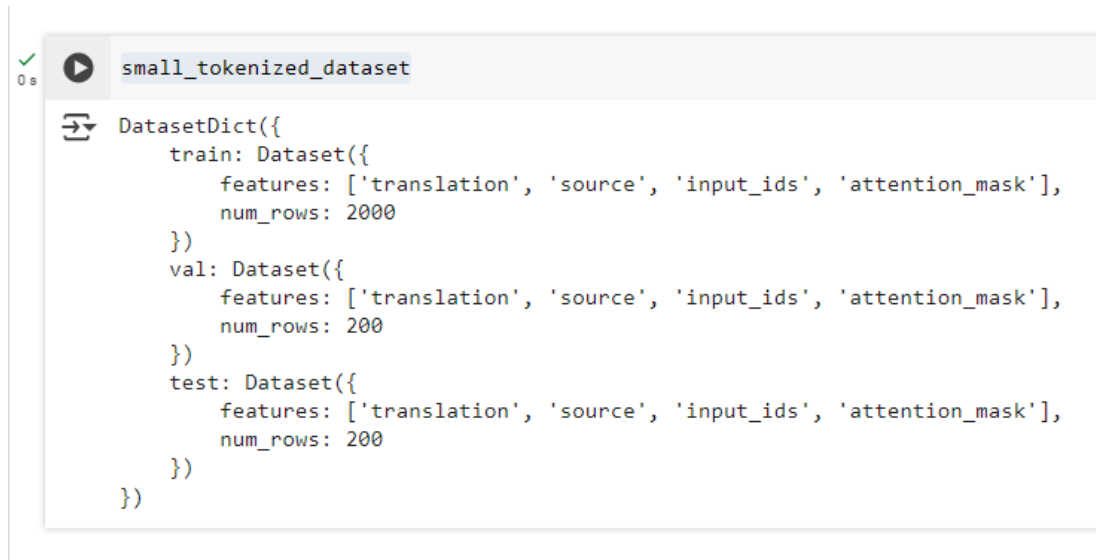
# Splitting dataset
small_be_en_dataset = DatasetDict(
    train = ds['train'].shuffle(seed=24).select(range(2000)).map(truncate),
    val = ds['train'].shuffle(seed=24).select(range(200)).map(truncate),
    test = ds['test'].shuffle(seed=24).select(range(200)).map(truncate),
)

# Tokenizing dataset
def tokenize_function(examples):
    return mt5_tokenizer(examples["source"], padding=True, truncation=True)

small_tokenized_dataset = small_be_en_dataset.map(tokenize_function, batched=True, batch_size=16)

data_collator = DataCollatorWithPadding(tokenizer=mt5_tokenizer)
```

Code 1: Splitting and preprocessing of the dataset for Model 1



```

0 s small_tokenized_dataset
DatasetDict({
  train: Dataset({
    features: ['translation', 'source', 'input_ids', 'attention_mask'],
    num_rows: 2000
  })
  val: Dataset({
    features: ['translation', 'source', 'input_ids', 'attention_mask'],
    num_rows: 200
  })
  test: Dataset({
    features: ['translation', 'source', 'input_ids', 'attention_mask'],
    num_rows: 200
  })
})

```

Figure 1: Splitting and preprocessing of the dataset for Model 1

This preprocessing method gave an opportunity to access source (marked as “source”) segments in BE and target (marked as “translation”) segments in EN to evaluate the performance in both directions separately.

For fine-tuning model 2, I have used another preprocessing method provided by the professor:

```

# Splitting dataset
from datasets import DatasetDict

small_dataset = DatasetDict(
    train = ds['train'].shuffle(seed=24).select(range(2000)),
    val = ds['validation'].shuffle(seed=24).select(range(200)),
    test = ds['test'].shuffle(seed=24).select(range(200)),
)

from transformers import AutoTokenizer

# Preprocessing dataset
source_lang = "be"
target_lang = "en"
prefix = "translate Belarusian to English: "
checkpoint = "google-t5/t5-small"
mt5_tokenizer = AutoTokenizer.from_pretrained(checkpoint)

def preprocess_function(examples):
    inputs = [prefix + example[source_lang] for example in examples["translation"]]
    targets = [example[target_lang] for example in examples["translation"]]
    model_inputs = mt5_tokenizer(inputs, text_target=targets, max_length=128, truncation=True)
    return model_inputs

```

```
tokenized = small_dataset.map(preprocess_function, batched=True)
```

Code 2: Splitting and preprocessing of the dataset for Model 2

```
0s [62] print(tokenized)

DatasetDict({
  train: Dataset({
    features: ['translation', 'input_ids', 'attention_mask', 'labels'],
    num_rows: 2000
  })
  val: Dataset({
    features: ['translation', 'input_ids', 'attention_mask', 'labels'],
    num_rows: 200
  })
  test: Dataset({
    features: ['translation', 'input_ids', 'attention_mask', 'labels'],
    num_rows: 200
  })
})
```

Figure 2: Splitting and preprocessing of the dataset for Model 2

This method allows to provide whole segments pairs line by line without specification of translation direction, as I needed to fine-tune Model 2 to perform the downstream task of translation in both directions.

Fine-Tuning Model 2

For fine-tuning Model 2, I experimented four times with manually tuning hyperparameters combinations, from which the most successful attempt was the last one:

```
training_args = Seq2SeqTrainingArguments(
    output_dir="/content/drive/MyDrive/fine_tuned_mt5_be_en_translation_model",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=5e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    weight_decay=0.05,
    #save_total_limit=3,
    num_train_epochs=7,
    predict_with_generate=True,
    fp16=True, #change to bf16=True for XPU
    report_to='none',
    seed=224
)
```

Code 3: Hyperparameter combination from attempt 4 of fine-tuning Model 2



Figure 3: Results from attempt 4 of fine-tuning Model 2

The previous attempts with their hyperparameter combinations and results can be seen in the Supplements.

Setup

I have conducted the whole project workflow using Google Collab with CPU. The checkpoints from the fine-tuning of Model 2 were stored in Google Collab and in Google Drive, while the best performing model (checkpoint-375) was stored additionally locally on my PC. This model was then used evaluation in zero-shot manner and for visualization of hidden states in TensorFlow.

Results

Quantitative results

For evaluating both models I have chosen 2 metrics, including Google BLEU and SacreBLEU:

- **Google BLEU** is said to be more effective on a sentence level than BLEU score. This metric can take on values from 0 to 1, inclusive. Higher scores are better, with 0 indicating no matches, and 1 indicating a perfect match.
- **SacreBLEU** is said to be more easily reproducible than just BLEU. The score can take any value between 0.0 and 100.0, inclusive, where 100.0 is a perfect match.

On the previous draft versions of the project, I have also used BLEU score, which is the convention metric in machine translation. BLEU's output is always a number between 0 and 1, where values closer to 1 represent more similar texts.

However, this metrics did not perform well on downstream task evaluation of Model 1 and had mostly zero's (the worst result) even on those segments, which were translated according to my manual observation partially or completely correctly. Therefore, I commented all the lines of code where it occurs and did not use it for the either evaluation Mode l1 and or for fine-tuning Model 2.

As it was mentioned before, I used three types of downstream tasks for translation in both directions separately: One sentence, a three sentence set as well as the validation set of for each model accordingly preprocessed dataset, printing and visualizing only its 15 segments.

For evaluation on a sentence level, I used the following sentences, where **en_be_reference** and **be_en_reference**, **mt5_en_be_sentence** and **mt5_be_en_sentence**, as well as **en_be_sentences** and **be_en_sentences** are the translations of each other and were used correspondingly for evaluation.

```
# Model 1 - One sentence
en_be_sentence = "<extra_id_6> Hello, how are you?"
en_be_reference = "Прывітанне, як справы?"
be_en_sentence = "<extra_id_2> Прывітанне, як справы?"
be_en_reference = "Hello, how are you?"

# Model 2 - One sentence
mt5_en_be_sentence = "Translate from English to Belarusian: Hello, how are you?"
mt5_be_en_sentence = "Translate from Belarusian to English: Прывітанне, як справы?"

# Three Sentence Set both for Model 1 and 2
en_be_sentences = [
    "I am learning programming.",
    "This is an amazing model!",
    "How is the weather today?"
]
be_en_sentences = [
    "Я вучуся праграмаваць.",
    "Гэта цудоўная мадэль!",
    "Як надвор'е сёння?"
]
```

Code 4: Reference and input sentences for one sentence and three sentence set evaluation for Model 1 and 2

Regarding the validation sets, the corresponding extracting syntax for values was used to access either BE or EN segments, e.g. indexing with “translation” in the preprocessed set for Model 1 and corresponding list comprehension with indexing “en” for Model 2.

```
#Model 1
val_set = small_tokenized_dataset["val"]["translation"]

#Model 2
val_set_en = [entry["en"] for entry in tokenized["val"]["translation"]]
```

Code 5: Splitting Validation Sets by Language for Model 1 and 2

As Model 2 does not perform on downstream tasks but outputs only **<extra_id_0>**, no metric calculations were performed and therefore are not presented in the further tables of results.

One Sentence Analysis

Model 1

One Sentence Scores - Model 1 (EN-BE):	One Sentence Scores - Model 1 (BE-EN):
Google BLEU = 0.29 SacreBLEU = 23.64	Google BLEU = 0.17 SacreBLEU = 11.51

Table 1: One Sentence Scores for Model 1

Model 2 after Fine-tuning

One Sentence Scores - Model 2 (EN-BE)	One Sentence Scores - Model 2 (BE-EN):
Google BLEU = 0.05 SacreBLEU = 4.03	Google BLEU = 0.0 SacreBLEU = 0.0

Table 2: One Sentence Scores for Model 2

From the scores, it is seen that Model 2 after fine-tuning still does not outperform Model 1, which was specifically trained for this downstream task. The higher scores tends to give SacreBLEU metrics, while Google BLEU score are much lower. For both models, the best performed translation direction is EN to BE, while the other was the worst for Mode 2 showing 0's in both metrics.

Three Sentence Set Analysis

Model 1

Three Sentence Set Scores - Model 1 (EN-BE)	Three Sentence Set Scores - Model 1 (BE-EN)
Sentence 1: Google BLEU = 0.11 SacreBLEU = 9.65	Sentence 1: Google BLEU = 0.08 SacreBLEU = 6.57
Sentence 2: Google BLEU = 0.4 SacreBLEU = 35.36	Sentence 2: Google BLEU = 0.27 SacreBLEU = 18.58
Sentence 3: Google BLEU = 0.4 SacreBLEU = 22.59	Sentence 3: Google BLEU = 0.82 SacreBLEU = 80.91

Table 3: Separate Three Sentence Set Scores for Model 1

Model 2 after Fine-tuning

Three Sentence Set Scores - Model 2 (EN-BE)	Three Sentence Set Scores - Model 2 (BE-EN)
Sentence 1: Google BLEU = 0.1 SacreBLEU = 15.97	Sentence 1: Google BLEU = 0.07 SacreBLEU = 0.0
Sentence 2: Google BLEU = 0.06 SacreBLEU = 8.12	Sentence 2: Google BLEU = 0.06 SacreBLEU = 0.0
Sentence 3: Google BLEU = 0.06	Sentence 3: Google BLEU = 0.04

SacreBLEU = 8.12	SacreBLEU = 5.52
------------------	------------------

Table 4: Separate Three Sentence Set Scores for Model 2

From these scores, it is seen that Model 2 after fine-tuning does not have such high scores for some sentences as Model 1 does, e.g. in sentence 2 and sentence 3 in both directions. However, Model 2 outperformed Model 1 in SacreBLEU in sentence 1 in EN to BE direction, while its Google BLEU score is close to zero.

The best result in both metrics was presented by Model 1 in BE to EN direction of sentence 3 (0.82 and 80.91), which is the tendency of Model 1 to perform better in EN to BE direction. However, this tendency was again stated by Model's 2 performance, where the opposite direction got scores close to zero.

Validation Set Analysis

Model 1

Overall Scores for the Whole Validation Set - Model 1 (EN-BE)	Overall Scores for the Whole Validation Set - Model 1 (BE-EN)
Google BLEU = 0.22 SacreBLEU = 19.03	Google BLEU = 0.17 SacreBLEU = 13.79

Table 5: Overall Validation Set Scores for Model 2

From the overall scores on validation set, it is seen that Model 1 performs better in direction EN to BE, while SacreBLEU scores are much higher than Google BLEU scores.

I think the scores are that low because the dataset itself is not absolutely "clean" as some source segments in Belarusian contain English words, only numbers or are unparallelled (See Code 5).

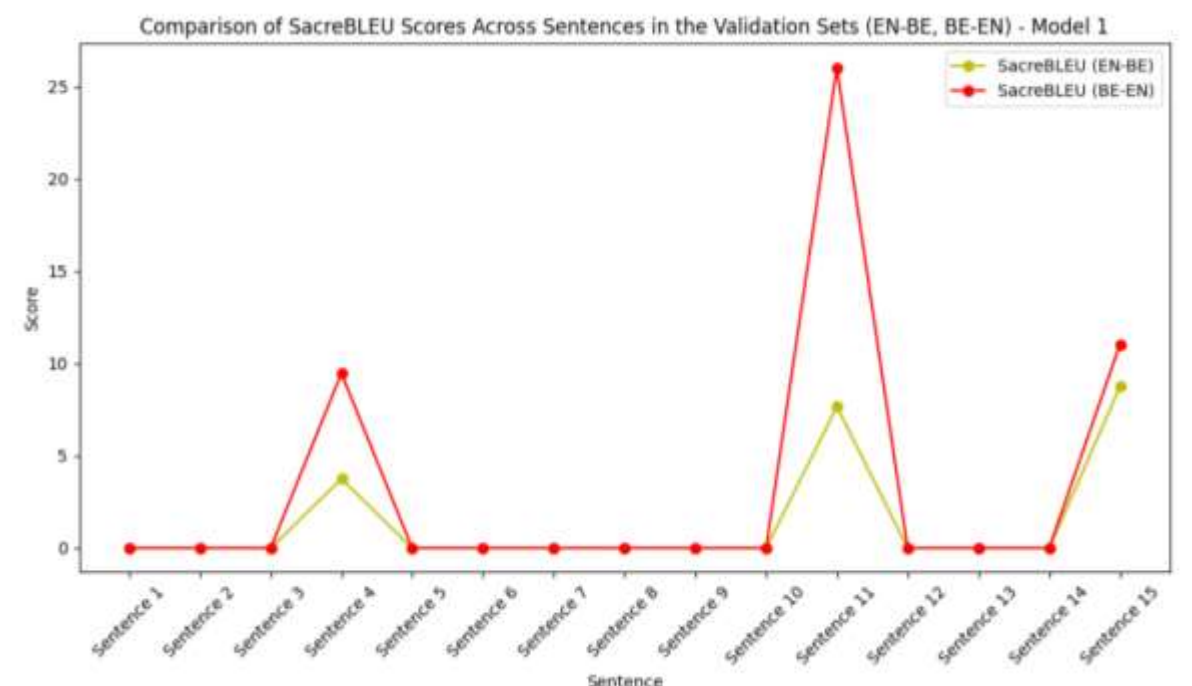


Figure 4: SacreBLEU Scores on 15 Segments for Validation Set for Model 1

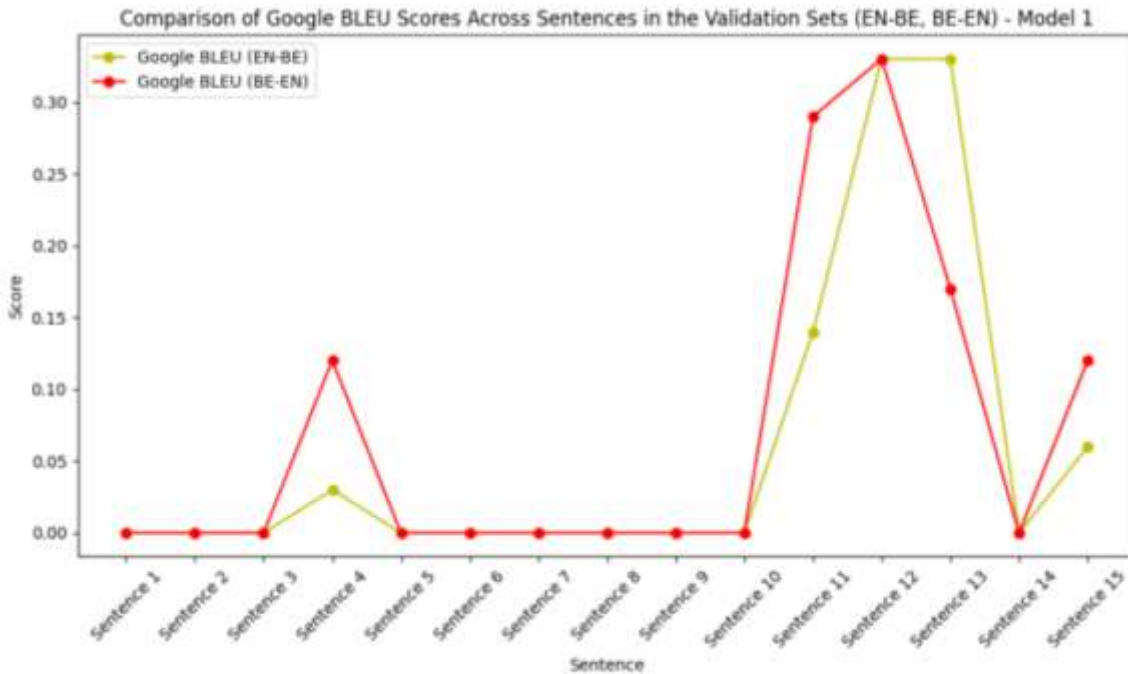


Figure 5: Google BLEU Scores on 15 Segments for Validation Set for Model 1

The illustrations show rather low scores for the 15 segments used from the validation set, where the SacreBLEU and Google BLEU show similar behavior (the latter showed more sentences with higher scores) with no significant difference in performance depending on the translation direction. There are also no segments that were translated with the highest values in both scores, e.g. sentences 12 and 13 have the highest Google BLEU scores, having the lowest in SacreBLEU.

This behavior also corresponds to the low overall SacreBLEU and Google BLEU scores for the whole validation set (See Table 5).

Model 2 after Fine-tuning

Overall Scores for the Whole Validation Set - Model 2 (EN-BE):	Overall Scores for the Whole Validation Set - Model 2 (BE-EN):
Google BLEU = 0.45 SacreBLEU = 45.88	Google BLEU = 0.04 SacreBLEU = 4.2

Table 6: Overall Validation Set Scores for Model 2

Surprisingly, the SacreBLEU score in direction EN to BE for the whole validation set outperformed more than twice the one of Model 1 (See Table 5), although Google BLEU and SacreBLEU in the other direction are significantly lower than the ones of Model 1. This behavior can assume the instable performance of Model 2, while Model 1 can perform more robustly.

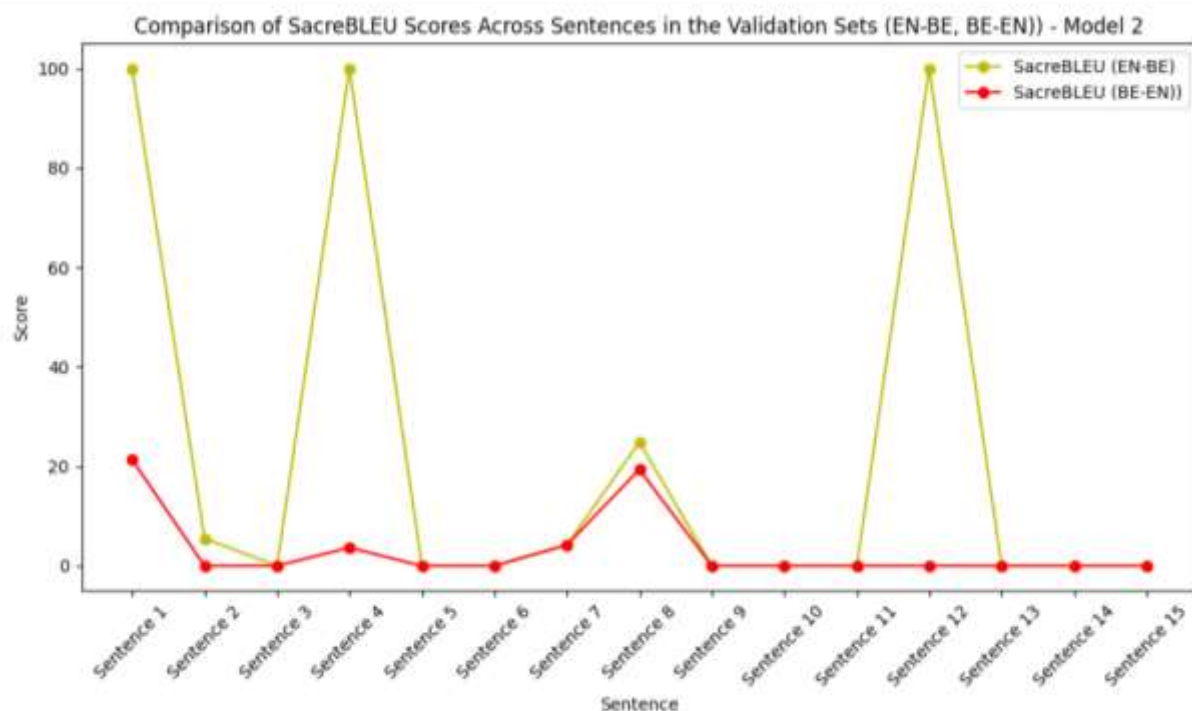


Figure 6: SacreBLEU Scores on 15 Segments for Validation Set for Model 2

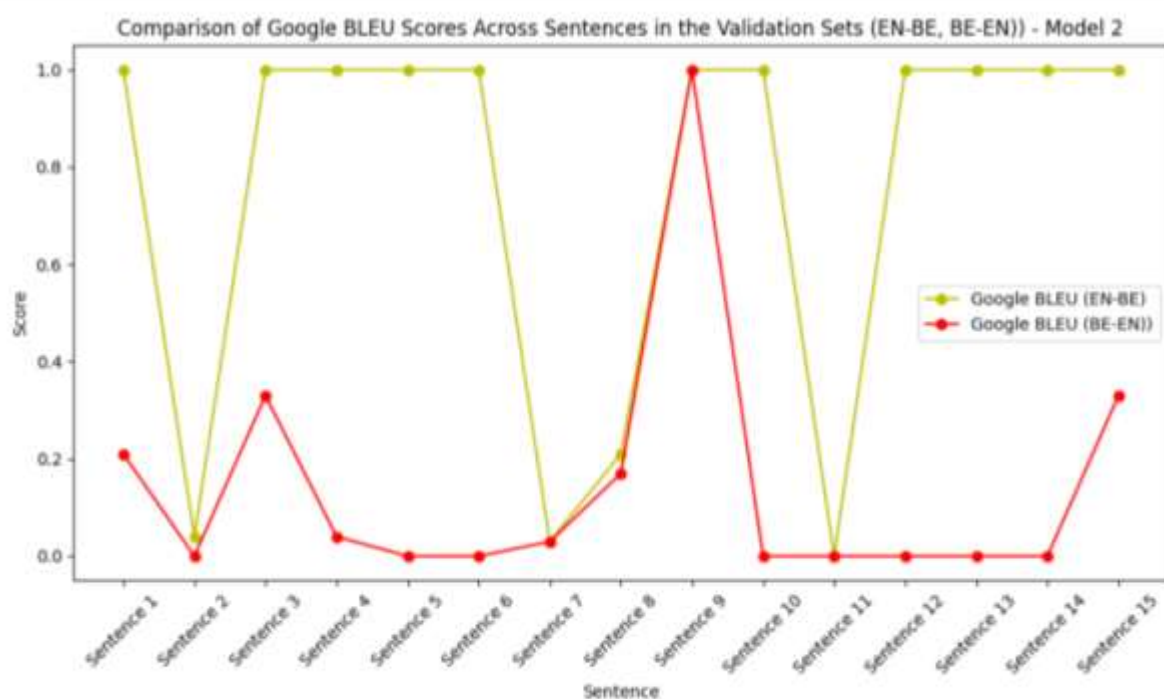


Figure 7: SacreBLEU Scores on 15 Segments for Validation Set for Model 2

This instability in performance of Model 2 can be also seen in the visualisations, where 11 sentences out of 15 were translated “perfectly” according to SacreBLEU score, while only 3 of these 11 (sentences 1, 4 and 12) got the same highest score in Google BLEU. The tendency of Model 2 better performing in EN to BE translation direction is also visible here (in yellow).

Test Set

Additionally, I also evaluated the performance of Model 2 on the whole test set, where only overall scores were taken into consideration.

Overall Scores for the Whole Test Set - Model 1 (EN-BE)	Overall Scores for the Whole Test Set - Model 1 (BE-EN)
Google BLEU = 0.27 SacreBLEU = 29.25	Google BLEU = 0.04 SacreBLEU = 3.9

Table 7: Overall Test Set Scores for Model 2

This table reflects similar behavior of Model 2 on validation set, giving another evidence for the previously observed tendency.

Qualitative results

One Sentence Analysis

Model 1

One Sentence Scores - Model 1 (EN-BE)	One Sentence Scores – Model 1 (BE-EN)
(Input) Hello, how are you? Translated Text: Hello, як вы? Reference: Прывітанне, як справы?	(Input) Прывітанне, як справы? Translated Text: Hello how to do? Reference: Hello, how are you?

Table 8: One Sentence Qualitative Analysis for Model 1

The EN input sentence was translated partially, leaving the first English word untranslated (“Hello”), while the BE input sentence was translated completely, but incorrectly (“How to do?” should have been “How are you?” or “How do you do?”)

Model 2 after Fine-tuning

One Sentence Scores - Model 2 (EN-BE)	One Sentence Scores – Model 2 (BE-EN)
(Input) Translate from English to Belarusian: Hello, how are you? Translated Text: Translate from English to Belarusian: Hello, how are you? Reference: Прывітанне, як справы?	(Input) Translate from Belarusian to English: Прывітанне, як справы? Translated Text: Reference: Hello, how are you?

Table 9: One Sentence Qualitative Analysis for Model 2

Here it is visible that translation direction impacts the output of Model 2 model as in direction BE to EN there is output, while in direction EN to BE it has one, although it is not a translation of the input sentence but its entire copy including prompt.

Three Sentence Set Analysis

Model 1

Model 1 (EN-BE)	Model 1 (BE-EN)
Original: I am learning programming. Translated: Я навучаю праграмаванне...	Original: Я вучуся праграмаваць. Translated: : I learn to program...

Original: This is an amazing model! Translated: Гэта дзіўная мадэль!	Original: Гэта цудоўная мадэль! Translated: : This is a wonderful model!
Original: How is the weather today? Translated: Як сёння надвор'е?	Original: Як надвор'е сёння? Translated: : How is the weather today?

Table 10: Three Sentence Set Qualitative Analysis for Model 1

In this table, it can be observed that some of the output sentences also contain different punctuation signs, e.g. in the translation of the first sentences of the Belarusian and English set respectively (one dot to three dots).

Regarding the translations, all of the sentences were translated correctly, using the same parts of speech (noun "programming" as noun "праграмаванне", while verb "праграмаваць" as verb "to program" (American English)) or synonyms ("wonderful" as "amazing" and "дзіўная" as "цудоўная").

Analyzing the corresponding scores (see Table 3), I would not score some of the translations as a human so low as they are complete and transmit the meaning correctly.

Model 2 after Fine-tuning

Model 2 (EN-BE)	Model 2 (BE-EN)
Original: I am learning programming. Translated: Ich lerne Programmierung.	Original: Я вучуся праграмаваць. Translated: .
Original: This is an amazing model! Translated: Das ist ein erstaunliches Modell!	Original: Гэта цудоўная мадэль! Translated: !
Original: How is the weather today? Translated: Wie ist das Wetter heute?	Original: Як надвор'е сёння? Translated: : How is the weather today?

Table 11: Three Sentence Set Qualitative Analysis for Model 2

Surprisingly, translation of three sentence set EN to BE revealed the possibility that Model 2 uses German as a pivot language, as all of the sentences were translated correctly but into the wrong language. In the other translation direction BE to EN, Model 2 provided only punctuation marks and a wrong translation for the last sentence.

Here, I am showing the 15 aligned segments from the datasets respectively for Model 1 and 2, which were used as references for metric calculations:

#Model 1

```
{'translation': ['Auto Away',
'Pas',
'Yearly',
'Insert Files - Insert new files into the project.',
'Line',
'1.600',
'Last',
'Fish Net',
```

```
'Drizzle Fog/ Mist',
'Select Message with Focus',
'This application is currently in offline mode. Do you want to connect in order to carry out this operation?',
'Level:',
'Trinidad And Tobago',
'Open at the given position in the bookmarks file',
'Fly your ship in a 4d environment'],
'source': ['Аўтаматычная адсутнасць',
'Сту',
'Штогод@ title: group',
'Даданне файлаў у праект',
'Лінейны',
'1, 600',
'Апошнія',
'Зрабіць фонавым малюнкамName',
'weather condition',
'Пазначыць & тэкст паведамлення',
'Праграма зараз у аўтаномным рэжыму. Ці вы хочаце злучыцца, каб выканаць гэтае дзеянне?',
'Узровень:',
'Трынідад і Табага',
'Стварыць тэчку...',
'Кіруйце касмічным караблём у чатырохмернай прасторы']}]
```

Code 6: Printed 15 Segments from Validation Set for Model 1

Model 2

```
[{'be': 'Гісторыя url.', 'en': 'History of combo url.'},
{'be': "Запамінаць назіраныя аб'екты паміж сесансамі", 'en': 'Remember the status of the tree.'},
{'be': 'Каментар:', 'en': 'Contact:'},
{'be': 'Ідэнтыфікатар ўліковага запісу тэчак IMAP.', 'en': 'This is the ID of the account holding the IMAP re-source folders.'},
{'be': 'Хуткасць:', 'en': '& Speed'},
{'be': 'name of the hash algorithm', 'en': 'SHA-512'},
{'be': 'З гэтым тэлескопам можна ўбачыць зоры і мары.', 'en': 'With this telescope stars and dreams can be seen.'},
{'be': '% 1 the hour translated above', 'en': 'twenty five to %1'},
{'be': 'keycombo', 'en': 'keycombo'},
```

```
{'be': 'Выканаўца', 'en': 'Artist'},
{'be': 'Фільтр апрацоўкі малюнкаComment', 'en': 'Filter effect definition'},
{'be': 'Знайсці папярэдні раз, калі сустракаецца пашуковы запыт', 'en': 'Find the previous match for the current search phrase'},
{'be': 'Нумар', 'en': 'Number'},
{'be': 'Не зважаць на сімвалічныя спасылкі', 'en': 'Discard all symlinks'},
{'be': 'Id сістэмы:', 'en': 'System Id:']}
```

Code 7: Printed 15 Segments from Validation Set for Model 2

It must be noticed that not all of the segments in both dataset sets are aligned properly, leaving some segments untranslated or even unparallelled. This characteristic of the dataset must have a strong influence on the output, however research of a more “clean” dataset for this particular language pair was no longer possible due to the time limit of this project.

Validation Set Analysis

Model 1

Model 1 (EN-BE)	Model 1 (BE-EN)
Original: Auto Away Translated: Auto Away	Original: Аўтаматычная адсутнасць Translated: Motormatic absence
Original: Pas Translated: Пас	Original: Сту Translated: Stu
Original: Yearly Translated: Годна	Original: Штогод@ title: group Translated: Every year@ title: group
Original: Insert Files - Insert new files into the project. Translated: Інсерт Філіс - Інсерт новыя файлы ў праект...	Original: Даданне файлаў у праект Translated: The addition of files to the project
Original: Line Translated: Лінія	Original: Лінейны Translated: Lense
Original: 1.600 Translated: медыятэхнічны і інш	Original: 1, 600 Translated: 1, 600
Original: Last Translated: Мінулае	Original: Апошнія Translated: The last
Original: Fish Net Translated: Рыба Net	Original: Зрабіць фонавым малюнкамName Translated: To make a background drawing ofName
Original: Drizzle Fog/ Mist Translated: Друзье Фог/місты	Original: weather condition Translated: weather condition
Original: Select Message with Focus Translated: Адбіраць Пасланне з Focus	Original: Пазначыць & тэкст паведамлення Translated: The information text will be considered

Original: This application is currently in offline mode. Do you want to connect in order to carry out this operation? Translated: Гэтае прыкладанне зараз у афлайн-рэжыме... Жадаешся падключыцца, каб правесці гэтую аперацыю?	Original: Праграма зараз у аўтаномным рэжыму. Ці вы хочаце злучыцца, каб выканаць гэтае дзеянне? Translated: The program is now in the autonomous regime... Do you want to connect to fulfill this effect?
Original: Level: Translated: Level:	Original: Узровень: Translated: Consection:
Original: Trinidad And Tobago Translated: Трынідад і Табара	Original: Трынідад і Табара Translated: Trinidad and Taba
Original: Open at the given position in the bookmarks file Translated: Адкрыцца на дадзенай пасадзе ў файце закладак	Original: Стварыць тэчку... Translated: Keep a leak...
Original: Fly your ship in a 4d environment Translated: Налётайце свой карабель у 4d-сучасце	Original: Кіруйце касмічным караблём у чатырохмернай прасторы Translated: Requit a space caral in a four-dimensional space

Table 12: Validation Set Qualitative Analysis for Model 1

In this table, it can be seen that in many cases different segments or even parts of words are used depending on the translation direction, e.g. Drizzle Fog/ Mist and weather condition. If there identical segments are used, these are mostly translated differently, e.g. Лінейны to Sense and Line to Лінія (the correct translation is the second one).

Model 2 after Fine-tuning

Model 2 (EN-BE)	Model 2 (BE-EN)
Original: History of combo url. Translated: History of combo url.	Original: Гісторыя url. Translated: url.
Original: Remember the status of the tree. Translated: Denken Sie an den Status des Baums.	Original: Запамінаць назірання аб'екты паміж сеансамі Translated: Delete the Deleted File
Original: Contact: Translated: Contact:	Original: Каментар: Translated: &
Original: This is the ID of the account holding the IMAP resource folders. Translated: This is the ID of the account holding the IMAP resource folders.	Original: Ідэнтыфікатар уліковага запісу тэчак IMAP. Translated: IMAP is a unified IMAP.
Original: & Speed Translated: & Speed	Original: Хуткасць: Translated: &
Original: SHA-512 Translated: SHA-512	Original: name of the hash algorithm Translated: Name of the hash algorithm
Original: With this telescope stars and dreams can be seen. Translated: Mit diesem Teleskop können Sterne und Träume gesehen werden.	Original: З гэтым тэлескопам можна ўбачыць зоры і мары. Translated: The symbols are not a problem.

Original: twenty five to %1 Translated: 25 bis %1	Original: % 1 the hour translated above Translated: %1 the hour translated above
Original: keycombo Translated: keycombo	Original: keycombo Translated: keycombo
Original: Artist Translated: Artist	Original: Выканаўца Translated: & &
Original: Filter effect definition Translated: Filter-Effekt Definition	Original: Фільтр апрацоўкі малюнкаComment Translated: Messages
Original: Find the previous match for the current search phrase Translated: Find the previous match for the current search phrase	Original: Знайсці папярэдні раз, калі сустракаецца па- шуковы запыт Translated: Default settings for the selected settings
Original: Number Translated: Number	Original: Нумар Translated: &
Original: Discard all symlinks Translated: Discard all symlinks	Original: Не зважаць на сімвалічныя спасылкі Translated: Default settings
Original: System Id: Translated: System Id:	Original: Ід сістэмы: Translated: Id:

Table 13: Validation Set Qualitative Analysis for Model 2

Visualization of Hidden States of Model 2

As **google/mt5-small** is an encoder-decoder model, I had to specify the hidden states of which part I going to extract from the fine-tuned Model 2 to visualize. In my project, I decided to visualize the layers 1 and 6 of the decoder, as it is focused more on generating the output than encoder.

Layer 1

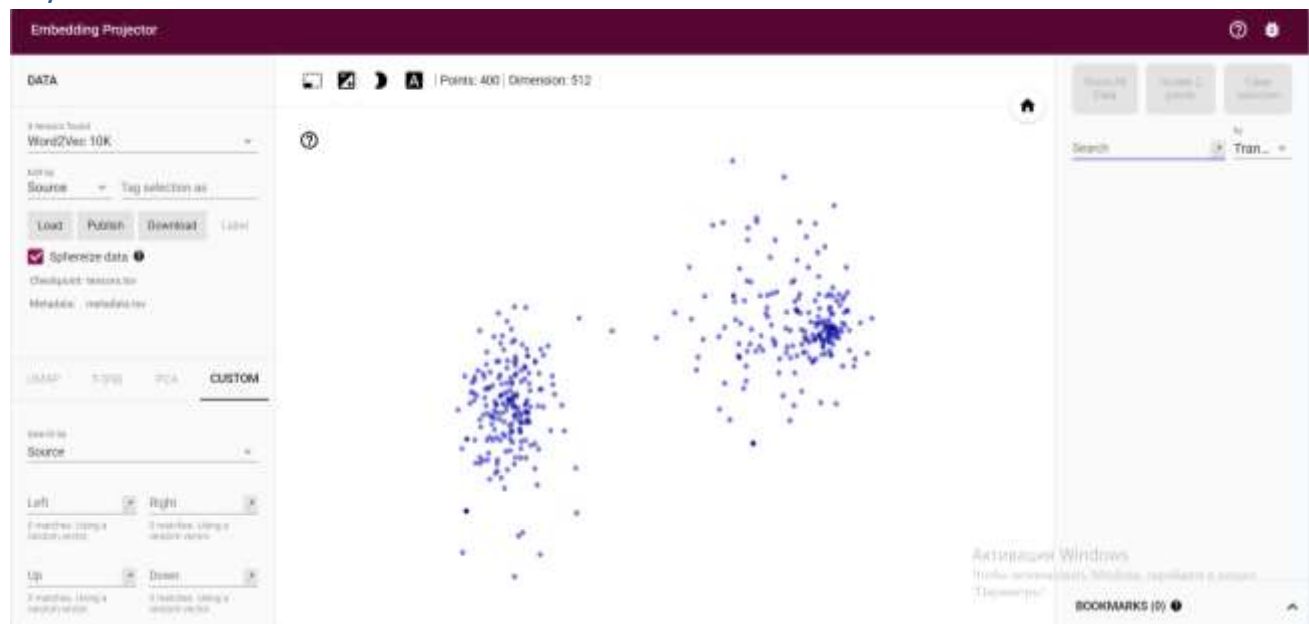


Figure 8: Decoder Hidden States – Layer 1



Figure 9: Decoder Hidden States – Layer 1 with One Selected Vector



Figure 10: Decoder Hidden States – Layer 1 with Nearest Neighbors of the Selected Vector

Layer 6

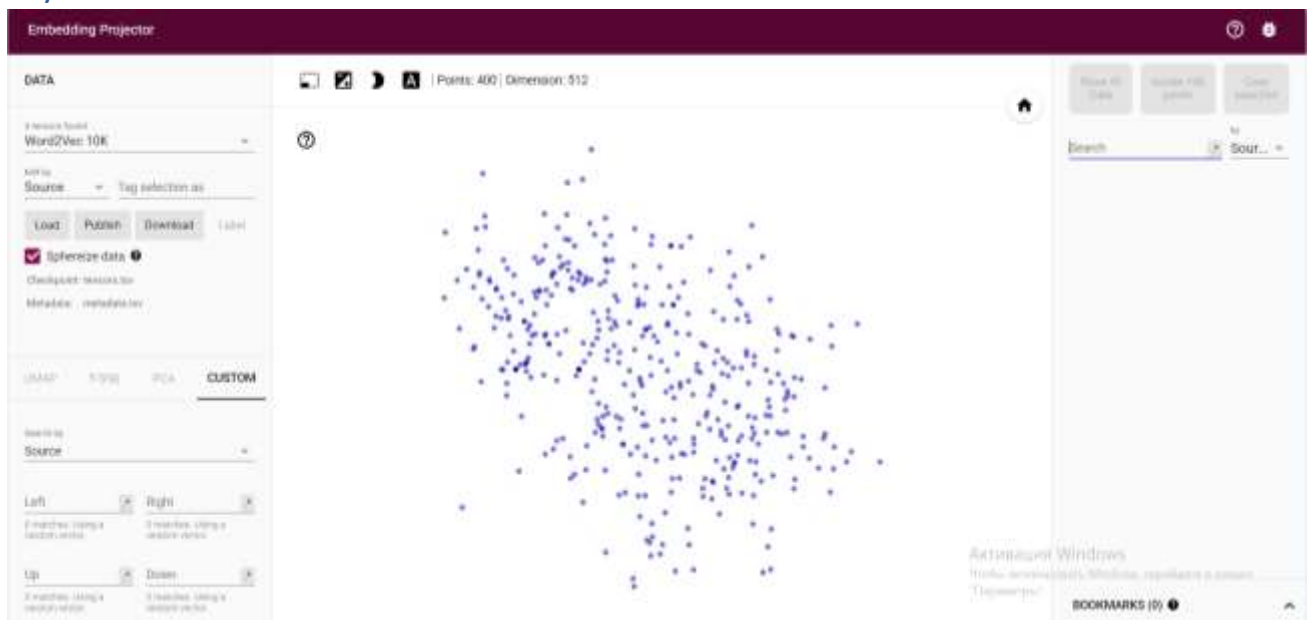


Figure 11: Decoder Hidden States – Layer 6



Figure 12: Decoder Hidden States – Layer 6 with One Selected Vector

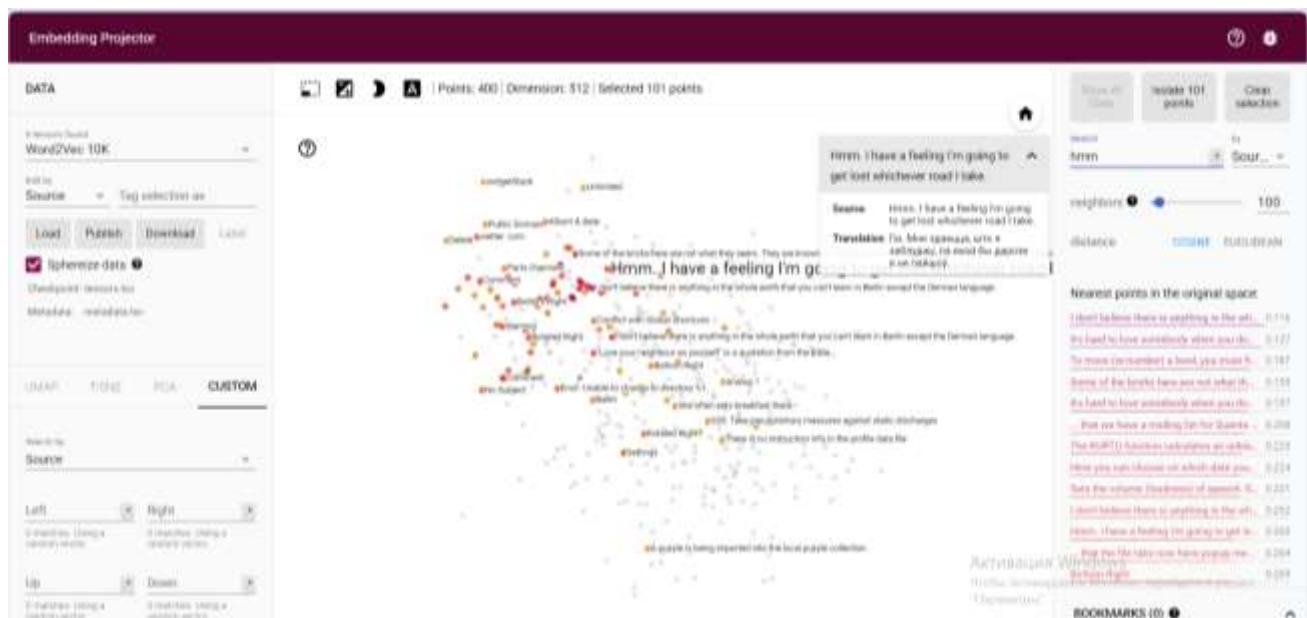


Figure 13: Decoder Hidden States – Layer 6 with Nearest Neighbors of the Selected Vector

Comparing the decoder hidden states from layers 1 and 6 it can be seen that the vectors were firstly arranged separately in two groups and were getting closer to each other though the layers towards the last one of decoder, getting diffused more equally. It is also visible on taking the same vector in both layers: In the first layer they are placed in the opposite groups, while in the sixth they are placed close to each other.

Moreover, I find it interesting that most of the nearest neighbors, calculated with cosine similarity, of this vector have also remained its nearest neighbors through the layers, although they got higher numbers and their order has also slightly changed, e.g. with “It’s hard to love...” as number 3 on layer 1 (0.007) and as number 2 on layer 6 (0.137).

Final summary

In conclusion, it has been proved that fine-tuning pretrained Model 2 can bring it to perform on machine translation tasks and be used further in zero-shot manner as Model 1. Since the initial state of Model 2 does not give any interpretable output, only the output from the fine-tuned state was compared with output of Model 1.

Regarding performance of Model 1, across all three types of downstream tasks it has performed quantitatively rather purely, where usually SacreBLEU scores are higher than Google BLEU scores. Additionally, in the visualizations of translated 15 segments from the validation set did not give any evidence on the connection of translation direction and scores, i. e. in both directions Model 1 performed rather equally. Qualitative analysis showed that most translations have been completed within the language pair, while untranslated segments remained only in one of the languages, and there were no empty spaces or repetitive character sequences as outputs.

Regarding performance of Model 2 after fine-tuning, it is still much lower quantitatively than the output of Model 1, especially in translation direction BE to EN, where there are mostly the lowest scores. On the unique example of SacreBLEU scores on the whole validation set and test set in direction EN to BE, Model 2 outperformed Model 1 (Compare Tables 5, 6 and 7). Although, the qualitative analysis showed that the translations were rather correct, but in the wrong language (German) or entirely remained unchanged. Additionally, the scores of the test set evaluation of Model 2 must be compared with the respective evaluation of Model 1 to give some objective insights, what was not however a part of this project.

Qualitative analyzing the outputs of Model 2 in translation across all types of downstream tasks revealed a “weakness” of Model 2 in translation BE to EN, where in most cases there everything but correct translation, including empty outputs (See Table 9), only punctuation signs (See Table 11) or repetitive character sequence with no meaning as output (see Table 13). Some instances of this behavior can be also observed in the other translation direction, but in a fewer amount.

Interesting observation can be seen in the EN to BE translation of a three sentence set where each of the sentences was translated correctly but into the wrong language. This behavior can lead to an assumption that Model 2 uses German as a pivot language in translation, however this can also not be the case with another language pair, where both languages are high-resource, e.g. Russian and English. Unfortunately, proving this hypothesis could not be handled in this project due to time limits.

Personal experience

For me it was interesting to learn about some existing models with the chosen language pair as Belarusian is a low-resource language compared to e.g. Russian. Besides, it was interesting to see whether fine-tuning can solve the problem of Model 2, as it was not trained on translation downstream tasks, although it supports 101 languages and its bigger model, which it is based on, can be used in zero-shot manner for such tasks.

The most difficult part was the fine-tuning of Model 2, which was later successfully solved thanks to the help of the professor, as well as extracting hidden states of Model 2 for visualization in TensorFlow, as Model 2 is an encoder-decoder model and the Seq2Seq task differs a lot from the Classification task we did in the class.

Disappointing part of the project was the final output of Model 2 after fine-tuning as I had thought of the output to be better or at least equal to the output of Model 1. Nevertheless, this can be possibly improved with more training time, automatic hyperparameter tuning using e.g. Optuna and/or more data in the training set.

For future projects, I will choose other downstream tasks like text classification as they are more interpretable and can also be evaluated more objectively using metrics than translation downstream tasks, where focus in evaluation lies, in my opinion, more on qualitative and manual analysis to give objective insights than on BLEU

scores tending to lower or mislead the results. However, this issue could have been solved with better prepared and “cleaner” dataset than mine I had used in this project.

References

- Model 1: <https://huggingface.co/WelfCrozzo/T5-L128-belarusian>
- Model 2: <https://huggingface.co/google/mt5-small>
- Fine-tuning Dataset: <https://huggingface.co/datasets/Helsinki-NLP/opus-100/viewer/be-en>
- BLEU Score: <https://huggingface.co/spaces/evaluate-metric/bleu>
- SacreBLEU: <https://huggingface.co/spaces/evaluate-metric/sacrebleu>
- Google BLEU Score: https://huggingface.co/spaces/evaluate-metric/google_bleu
- mT5 Paper: <https://arxiv.org/pdf/2010.11934>
- mt5 Video: https://www.youtube.com/watch?v=xbLo6jt3_Bc
- C4 Training Dataset of mt5: <https://www.tensorflow.org/datasets/catalog/c4#c4multilingual>
- mt5 on AIModels.fyi: <https://www.aimodels.fyi/models/huggingFace/mt5-small-google>
- TensorFlow: <https://projector.tensorflow.org/>
- mt5 for MT: <https://stackoverflow.com/questions/76040850/can-mt5-model-on-huggingface-be-used-for-machine-translation>
- Fine-tuning Code Example: https://github.com/ejmejm/multilingual-nmt-mt5/blob/main/nmt_full_version.ipynb

Supplements

Attempt 1

```
training_args = Seq2SeqTrainingArguments(
    output_dir="fine_tuned_mt5_be_en_translation_model",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    weight_decay=0.01,
    #save_total_limit=3,
    num_train_epochs=3,
    predict_with_generate=True,
    fp16=True, #change to bf16=True for XPU
    report_to='none',
    seed=224
)
```

Code 8: Hyperparameter combination from attempt 1 of fine-tuning Model 2

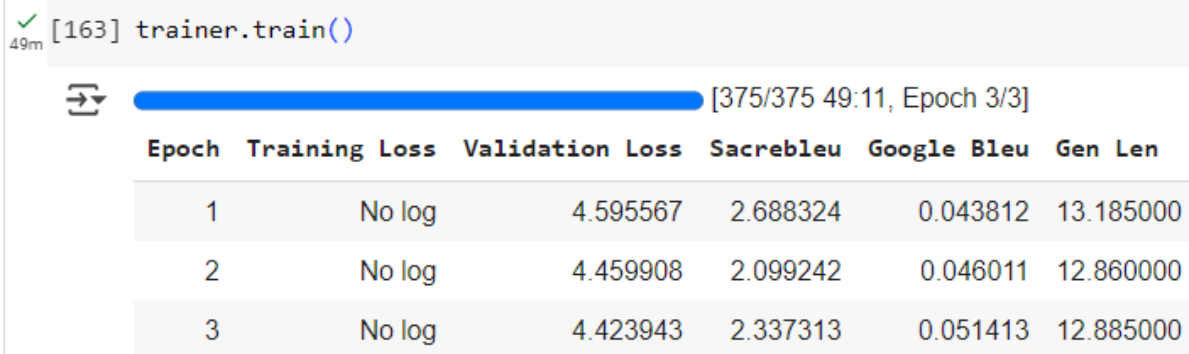


Figure 14: Results from attempt 1 of fine-tuning Model 2

Attempt 2

```
training_args = Seq2SeqTrainingArguments(
    output_dir="/content/drive/MyDrive/fine_tuned_mt5_be_en_translation_model",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=3e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    weight_decay=0.01,
    #save_total_limit=3,
    num_train_epochs=5,
    predict_with_generate=True,
    fp16=True, #change to bf16=True for XPU
    report_to='none',
    seed=224
)
```

Code 9: Hyperparameter combination from attempt 2 of fine-tuning Model 2

✓ 1h [80] trainer.train()

➡ Passing a tuple of `past\_key\_values` is deprecated and will be removed in [625/625 1:08:16, Epoch 5/5]

Epoch	Training Loss	Validation Loss	Sacrebleu	Google Bleu	Gen Len
1	No log	4.456567	2.148740	0.045565	12.995000
2	No log	4.308761	3.084865	0.057734	12.375000
3	No log	4.261287	3.666821	0.061734	12.130000
4	4.439100	4.239805	3.111321	0.053283	12.150000
5	4.439100	4.233558	3.088701	0.052694	12.250000

Figure 15: Results from attempt 2 of fine-tuning Model 2

Attempt 3

```
training_args = Seq2SeqTrainingArguments(
    output_dir="/content/drive/MyDrive/fine_tuned_mt5_be_en_translation_model",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=5e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    weight_decay=0.03,
    #save_total_limit=3,
    num_train_epochs=5,
    predict_with_generate=True,
    fp16=True, #change to bf16=True for XPU
    report_to='none',
    seed=224
)
```

Code 10: Hyperparameter combination from attempt 3 of fine-tuning Model 2

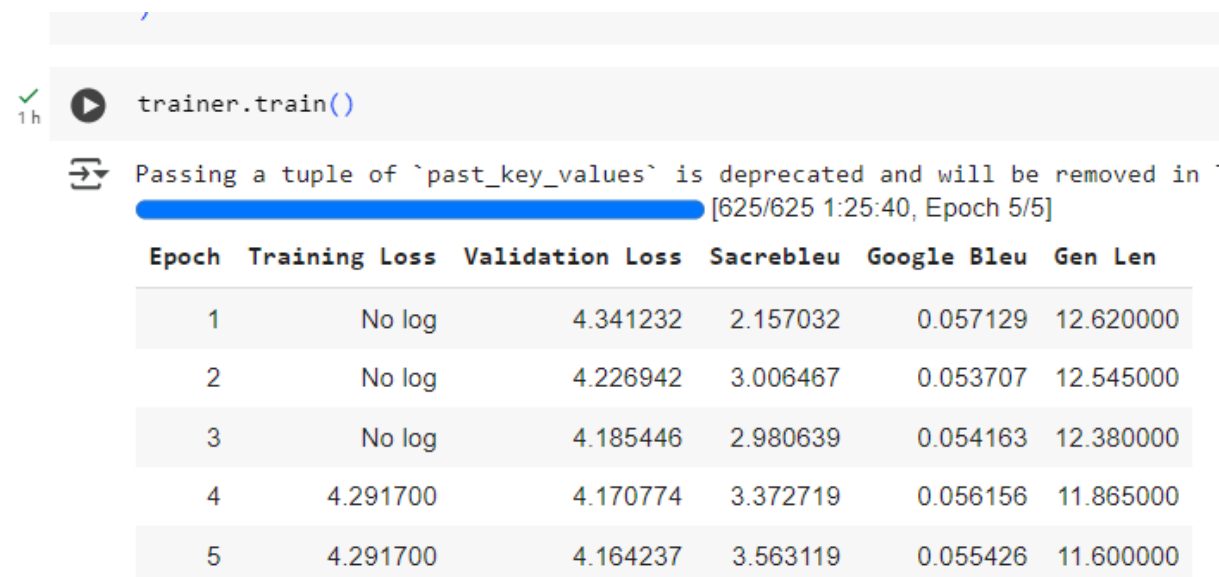


Figure 16: Results from attempt 3 of fine-tuning Model 2